```
! Config Multilayer Switch
enable
configure terminal
hostname MLS-INSIDE
no ip domain-lookup
! Tạo VLAN
vlan 10
 name USERS_10
exit
vlan 20
 name USERS_20
exit
! Gán cổng cho PC0 vào VLAN 10
interface fastEthernet0/5
 description PC0_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
exit
! Gán cổng cho PC1 vào VLAN 20
interface fastEthernet0/10
 description PC1_VLAN20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 no shutdown
exit
! Bật định tuyến và tạo SVI
(Gateway)
ip routing
interface vlan 10
 description GATEWAY_VLAN10
 ip address 192.168.10.1
255.255.255.0
 no shutdown
exit
interface vlan 20
 description GATEWAY_VLAN20
 ip address 192.168.20.1
255.255.255.0
 no shutdown
exit
! Config cổng Uplink lên ASA
thành Routed Port
interface gigabitEthernet0/1
 description
ROUTED_LINK_TO_ASA_INSIDE
 no switchport
 ip address 192.168.99.1
255.255.255.252
 no shutdown
exit

! Định tuyến mặc định đẩy dữ liệu
ra ASA
ip route 0.0.0.0 0.0.0.0
192.168.99.2
end
write memory
! Config Tường lửa Cisco ASA
5506-X (ASA-HQ)
enable
configure terminal
hostname ASA-HQ
domain-name ccna.lab
enable password Ciscoenpa55
passwd Ciscocopa55

! Config các vùng mạng (Interfaces)
interface gigabitEthernet1/1
 description OUTSIDE_TO_R0
 nameif outside
 security-level 0
 ip address 10.0.0.2 255.255.255.0
 no shutdown
exit
interface gigabitEthernet1/2
 description INSIDE_TO_MLS
 nameif inside
 security-level 100
 ip address 192.168.99.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet1/3
 description DMZ_TO_SERVERS
 nameif dmz
 security-level 50
 ip address 172.16.30.1 255.255.255.0
 no shutdown
exit
! Định tuyến tĩnh
route outside 0.0.0.0 0.0.0.0 10.0.0.1
route inside 192.168.10.0 255.255.255.0
192.168.99.1
route inside 192.168.20.0 255.255.255.0
192.168.99.1
! Config NAT (PAT) cho VLAN 10 và VLAN
20
object network NET_VLAN10
 subnet 192.168.10.0 255.255.255.0
 nat (inside,outside) dynamic interface
exit
object network NET_VLAN20
 subnet 192.168.20.0 255.255.255.0
 nat (inside,outside) dynamic interface
exit
! Config ACL cho phép Web và DNS từ
Outside vào DMZ
access-list OUTSIDE_IN extended permit
tcp any host 172.16.30.20 eq 80
access-list OUTSIDE_IN extended permit
tcp any host 172.16.30.20 eq 443
access-list OUTSIDE_IN extended permit
udp any host 172.16.30.10 eq 53
access-list OUTSIDE_IN extended permit
tcp any host 172.16.30.10 eq 53
access-group OUTSIDE_IN in interface
outside
! Config SSH chỉ cho phép VLAN 10 truy cập
username admin password Ciscosecure123
privilege 15
aaa authentication ssh console LOCAL
crypto key generate rsa modulus 1024
ssh 192.168.10.0 255.255.255.0 inside
ssh version 2
ssh timeout 5
! Config Stateful Inspection cho ICMP (cho
phép Ping)
policy-map global_policy
 class inspection_default
  inspect icmp
exit
service-policy global_policy global

end
write memory

! Config Router R0 (Kết nối ASA và OSPF)
enable
configure terminal
hostname R0
no ip domain-lookup
! Config IP cổng nối ASA và cổng mạng OSPF Transit
interface gigabitEthernet0/0
 description TO_ASA_OUTSIDE
 ip address 10.0.0.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet0/1
 description OSPF_TRANSIT
 ip address 10.10.10.1 255.255.255.248
 no shutdown
exit
! Định tuyến tĩnh hướng về ASA
ip route 172.16.30.0 255.255.255.0 10.0.0.2
ip route 192.168.10.0 255.255.255.0 10.0.0.2
ip route 192.168.20.0 255.255.255.0 10.0.0.2
! Config OSPF Process 10
router ospf 10
 router-id 1.1.1.1
 network 10.10.10.0 0.0.0.7 area 0
 network 10.0.0.0 0.0.0.255 area 0
 passive-interface gigabitEthernet0/0
 redistribute static subnets
exit
end
write memory
! Config Router R1 (Chi nhánh 1 - 100 hosts)
enable
configure terminal
hostname R1
! Config cổng
interface gigabitEthernet0/0
 description OSPF_TRANSIT
 ip address 10.10.10.2 255.255.255.248
 no shutdown
exit
interface gigabitEthernet0/1
 description BRANCH_100_HOSTS
 ip address 172.16.10.1 255.255.255.128
 no shutdown
exit
! Config OSPF
router ospf 10
 router-id 2.2.2.2
 network 10.10.10.0 0.0.0.7 area 0
 network 172.16.10.0 0.0.0.127 area 0
 passive-interface gigabitEthernet0/1
exit
end
write memory
! Config Router R2 (Chi nhánh 2 - 60 hosts)
enable
configure terminal
hostname R2
! Config cổng
interface gigabitEthernet0/0
 description OSPF_TRANSIT
 ip address 10.10.10.3 255.255.255.248
 no shutdown
exit
interface gigabitEthernet0/1
 description BRANCH_60_HOSTS
 ip address 172.16.20.1 255.255.255.192
 no shutdown
exit
! Config OSPF
router ospf 10
 router-id 3.3.3.3
 network 10.10.10.0 0.0.0.7 area 0
 network 172.16.20.0 0.0.0.63 area 0
 passive-interface gigabitEthernet0/1
exit
end
write memory
```